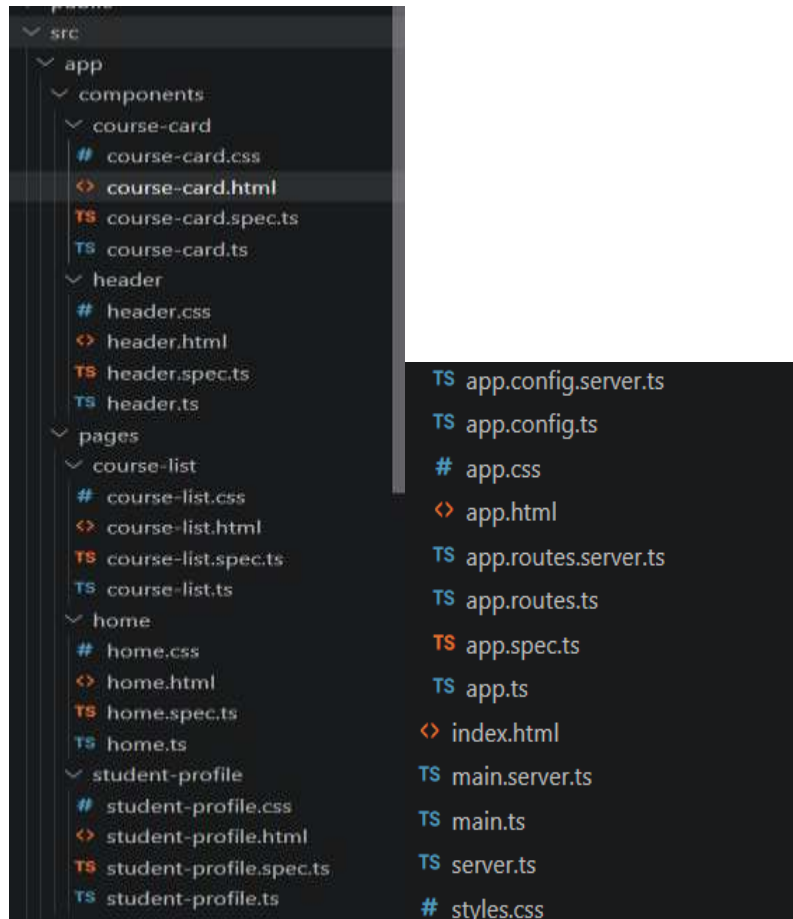


HANDS-ON 2 [Beginner] Data Binding, Lifecycle Hooks & Component Communication

Commands Used

- `ng new student-course-portal`
- `cd student-course-portal`
- `ng generate component components/header`
- `ng generate component pages/home`
- `ng generate component pages/course-list`
- `ng generate component pages/student-profile`
- `ng generate component components/course-card`
- `ng serve`

Project Structure



course-list.ts

```
import { Component } from '@angular/core';
```

```
import { CommonModule } from '@angular/common';
import { CourseCard } from '../components/course-card/course-card';
@Component({
  selector: 'app-course-list',
  standalone: true,
  imports: [CommonModule, CourseCard],
  templateUrl: './course-list.html',
  styleUrls: ['./course-list.css']
})
export class CourseList {
  courses = [
    {
      id: 1,
      name: 'Angular',
      code: 'ANG101',
      credits: 4
    },
    {
      id: 2,
      name: 'Java',
      code: 'JAVA201',
      credits: 3
    },
    {
      id: 3,
      name: 'Python',
      code: 'PY301',
      credits: 2
    }
  ]
}
```

```

    },
    {
      id: 4,
      name: 'Data Structures',
      code: 'DS401',
      credits: 4
    },
    {
      id: 5,
      name: 'Machine Learning',
      code: 'ML501',
      credits: 5
    }
  ];
  selectedCourseId: number = 0;
  onEnroll(courseId: number): void {
    this.selectedCourseId = courseId;
    console.log('Course ID:', courseId);
    console.log('Selected Course ID:', this.selectedCourseId);
  }
}

```

course-list.html

Displays course cards and listens to enrollRequested event.

```
<h2>Available Courses</h2>
```

```
<app-course-card
```

```
  *ngFor="let course of courses"
```

```
  [course]="course"
```

```

        (enrollRequested)="onEnroll($event)">
    </app-course-card>

    <hr>

    <p *ngIf="selectedCourseId > 0">
        <strong>Selected Course ID:</strong>
        {{ selectedCourseId }}
    </p>

```

course-card.ts

Uses @Input(), @Output(), EventEmitter and ngOnChanges().

```

import {
    Component,
    Input,
    Output,
    EventEmitter,
    OnChanges,
    SimpleChanges
} from '@angular/core';

@Component({
    selector: 'app-course-card',
    imports: [],
    templateUrl: './course-card.html',
    styleUrls: ['./course-card.css']
})

export class CourseCard implements OnChanges {
    @Input()
    course: any;

    @Output()
    enrollRequested = new EventEmitter<number>();

```

```

ngOnChanges(changes: SimpleChanges): void {
  console.log('Previous Value:', changes['course']?.previousValue);
  console.log('Current Value:', changes['course']?.currentValue);
}

enroll() {
  this.enrollRequested.emit(this.course.id);
}
}

```

course-card.html

Displays course details and Enroll button.

```

<div class="course-card">

  <h3>{{ course.name }}</h3>

  <p><strong>Code:</strong> {{ course.code }}</p>

  <p><strong>Credits:</strong> {{ course.credits }}</p>

  <button (click)="enroll()">

    Enroll

  </button>

</div>

```

Output

```

PS C:\Users\prava\student-course-portal> ng serve
Port 4200 is already in use.
Would you like to use a different port? Yes
Browser bundles:
Initial chunk files | Names | Raw size
main.js | main | 21.33 KB
styles.css | styles | 95 bytes
Initial total | 21.43 KB

Server bundles
Initial chunk files | Names | Raw size
main.server.mjs | main.server | 22.64 KB
server.mjs | server | 1.84 KB
polyfills.server.mjs | polyfills.server | 268 bytes

Application bundle generation complete. [6.670 seconds] - 2026-07-27T23:32:31.660Z

Watch mode enabled. Watching for file changes...
NOTE: Raw file sizes do not reflect development server per-request transformations.
→ Local: http://localhost:55548/
→ press h + enter to show help

```

